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Network Design & Implementation

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Introduction

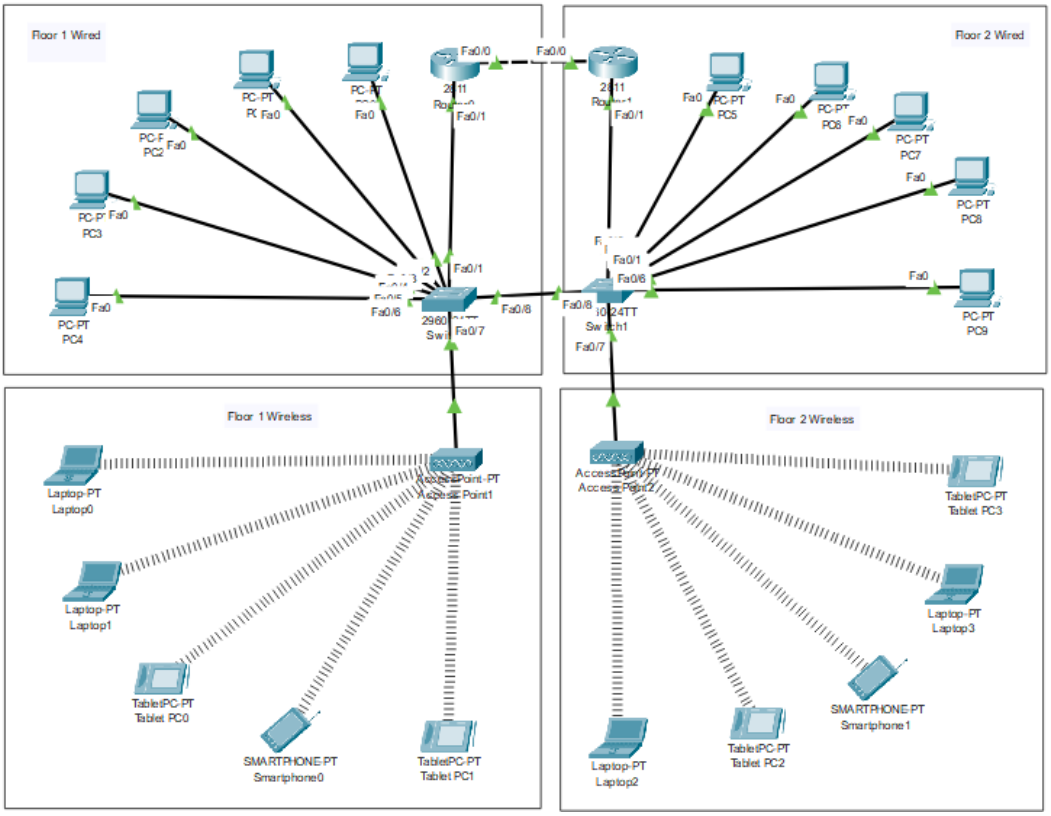
In this report, I will plan, design and implement a medium-level network for an advertising client that will need to accommodate around 50 employees that will need access. For this, I will have to create a CIDR IP addressing scheme to specify IP addresses for the deployed network devices to ensure its scalability across the network. This will also include my configuration for a Secure Shell setup on the routers to manage the network remotely. I will also display the connectivity results of the devices and how they are connected as well as verify RIP and DCHP configurations as well as other protocols included.

For this project, I will be using Cisco packet tracer to plan the network and all its devices in an offline environment which will also allow me to find the best solution to this request. This software will also allow me to set up and configure all the network devices properly, such as routers, switches and the computer devices themselves. The Cisco packet tracer will also allow me to run fake packet messages through the network to test the network works and how the information flows through the network and to each device.

Aim and objectives

My aim for this project is to produce the final network plan and implementation for the client, accommodating to all their needs. This will need to be fully functioning whilst using the most efficient methods and devices to provide a great outcome for the network. My objectives for this project are to provide a CIDR IP addressing scheme to allow scalability, Configure Secure shell, provide connectivity results for devices, demonstrate and verify RIP and DCHP configurations. Another objective throughout is to provide images of the progress and show configuration details and display the network diagram.

Network Design



The design of this network will consist of having two routers for each floor, which will connect to their floor's dedicated switch. This switch will grant the floor's devices to connect through the switch, to the router therefore giving it access to the internet. This will allow access for all 10 devices on each floor. Each switch will also have a wireless access point connected, this will give the clients mobile devices such as tablets, laptops and smartphones the options to also connect to the internet wirelessly on each floor.

This network will fulfil all the client's needs, allowing access to 20 desks to connect on each floor (40 in total), whilst also featuring a wireless access point for other devices to easily connect to the internet wirelessly. This design will also work very efficiently without an unnecessary amount of network devices, this was done by using a minimal amount to be able to meet the client's requirements and allow all devices a simple and efficient method of efficiency.

The objective of the design plan I created was to make a network that firstly meets the client's needs, includes a great configuration to boost its performance, security and efficiency as well as having measures in place for redundancy. To implement the redundancy devices the switches were also connected for any router outage so they could still communicate in this possible failure. For performance, I will use DHCP, RIP and SSH to allow for a lot more connectivity methods and failsafe throughout the network and for the scalability of the design.

With this network, I have chosen specific devices to ensure that this network is very reliable and to ensure scalability. Using 2960 switches allows for up to 24 devices to connect, whilst they currently need 20 per floor these allow for future ease of scalability by having extra ports to accommodate, as well as wireless connections for mobile devices and laptops etc.

This plan for the network will allow for it to work efficiently for the client as well as be able to host an overall network design for 40 wired connections and many wireless devices. With this, the network also has two separate routers in place to communicate through each floor, as well as having both floors' switches connected. This is in place in case one of the routers were to encounter problems or go offline.

For the devices across the network I used copper straight-through cables as these are the best to use for connecting PCs to a switch, and also the most recommended for connecting switches and routers. This, therefore, made more sense in the planning stage to use the copper straight-through cables for the whole of the design.

Configuration Details

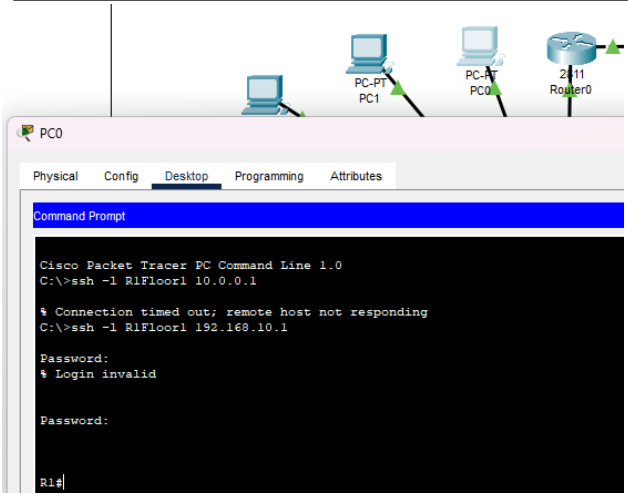
SSH

```
R1(config)#ip domain-name SSHR1
R1(config)#crypto key generate rsa general-keys modulus 512
% You already have RSA keys defined named R1.SSHRouter
% They will be replaced.

% The key modulus size is 512 bits
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 1:24:10.722: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 1:24:10.722: %SSH-5-ENABLED: SSH 1.5 has been enabled
R1(config)#username R1Floor1 privilege 15 password R1Pass
R1(config)#line vty 0 15
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
R1(config)#crypto key generate rsa general-keys modulus 768
% You already have RSA keys defined named R1.SSHR1
% They will be replaced.

% The key modulus size is 768 bits
% Generating 768 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 1:25:25.441: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#ip ssh version 2
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write
Building configuration...
[OK]
R1#
```



Using the Secure Shell protocol (SSH) allowed me to configure many devices to be able to work better

within the network, not only for efficiency but for other factors such as security, cost reduction, simpler management as well as better mitigation in the network. In the images above it shows me enabling SSH on Floor 1's Router, this allows the user devices to connect and run shell commands remotely on the router as soon above, connecting to Router 1 and being able to run commands on that router.

DHCP

Switch Configuration

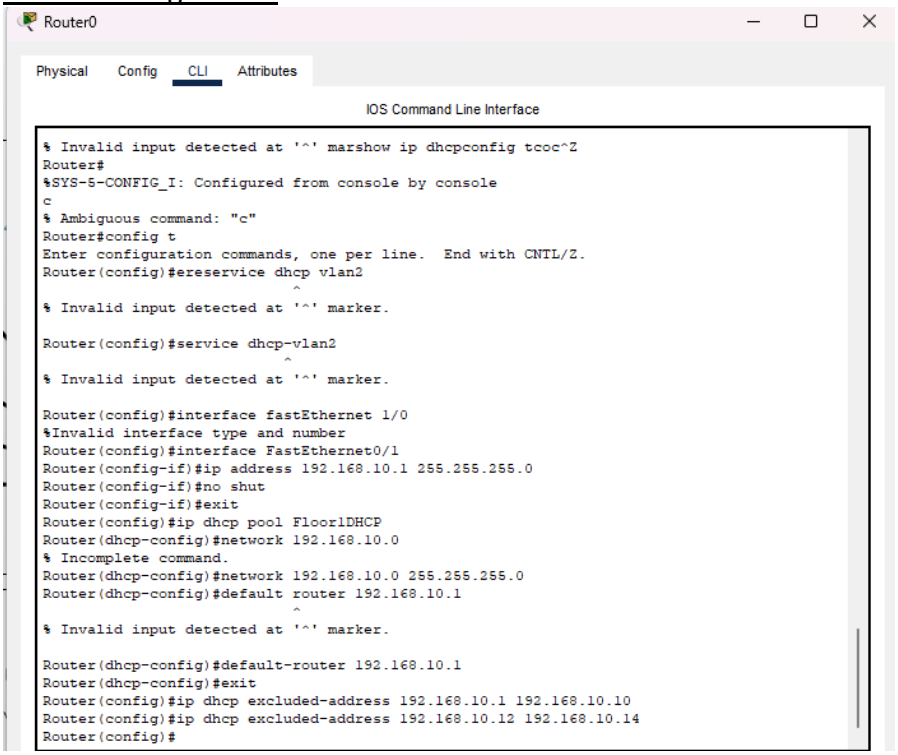
```
Switch(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.10
Switch(config)#ip dhcp pool test
Switch(dhcp-config)#network 192.168.1.0
% Incomplete command.
Switch(dhcp-config)#network 192.168.1.0 255.255.255.0
Switch(dhcp-config)#default-router 192.168.1.1
Switch(dhcp-config)#dns-server 4.4.4.4
Switch(dhcp-config)#exit
Switch(config)#interface vlan 1
Switch(config-if)#ip address 192.168.1.5 255.255.255.0
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

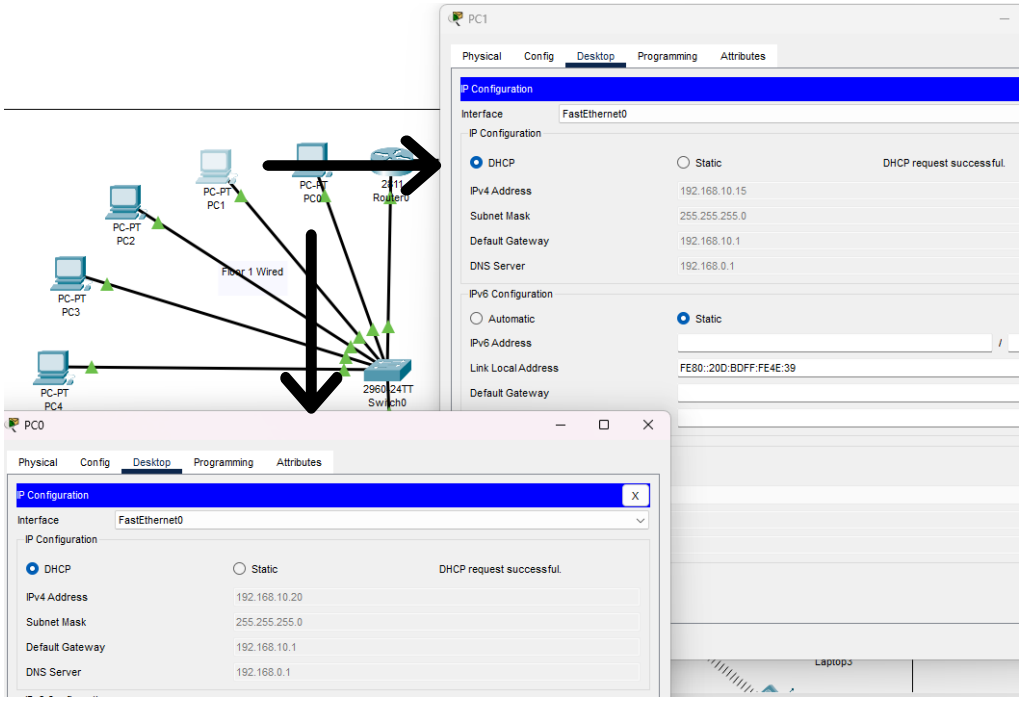
Switch(config-if)#no shutdown
Switch(config-if)#exit
```

Router Configuration



Above shows me configuring the DHCP server on the Switch & Router, by doing this it defines a DHCP pool of IP addresses that can be assigned to the devices, this also provides a default gateway for the LAN and a DNS server for the devices.

After this configuration on the Router, we can then go and enable DCHP on the devices that are connected to that router/switch, automatically assigning them IP addresses.



We then complete this for the second floors wired network to allow for that to also use DHCP also and then setup the computers to request for the DHCP IP configuration.

Once this has been done, we are able to ping for example PC0 -> PC1.

```
C:\>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

DHCP allows for the network to have a reliable IP address configuration, this is because it minimizes configuration errors that are commonly caused by having manual IP address configuration, for any errors such as more than one computer having the same IP address as another.

RIP

```
R1(config-router)#config t
%Invalid hex value
R1(config)#router rip
R1(config-router)#network 10.0.0.0
R1(config-router)#net
% Incomplete command.
R1(config-router)#network 192.168.1.0
R1(config-router)#network 20.0.0.0
R1(config-router)#exit
```

```
R2#config t
Enter configuration commands, one per l
R2(config)#router rip
R2(config-router)#network 10.0.0.0
R2(config-router)#network 192.168.1.0
R2(config-router)#network 20.0.0.0
R2(config-router)#exit
R2(config)#show ip route
```

```
Gateway of last resort is not set

R    10.0.0.0/8 [120/1] via 192.168.1.1, 00:00:51, FastEthernet0/0
C    20.0.0.0/8 is directly connected, FastEthernet0/1
C    192.168.1.0/24 is directly connected, FastEthernet0/0
```

"Distance vector routing algorithm assumes that each router maintains a table (e.g., a vector) that preserves the best-known distance to each destination and the line to be followed to get there, as in". (Ioan Fiţigău, 2013) This gives the packets a path to be able to get to its end destination in the network. The RIP uses a routing table which contains a list of the destinations for the router to understand how to reach each device.

The main advantages RIP offers are that it is easy to understand and to implement into the network as well as providing loop-free routing to avoid causing issues with the same packets ending up back at the same router.

CIDR IP Addressing Scheme

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external t
       E1 - OSPF external type 1, E2 - OSPF external type 2, E -
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - I
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

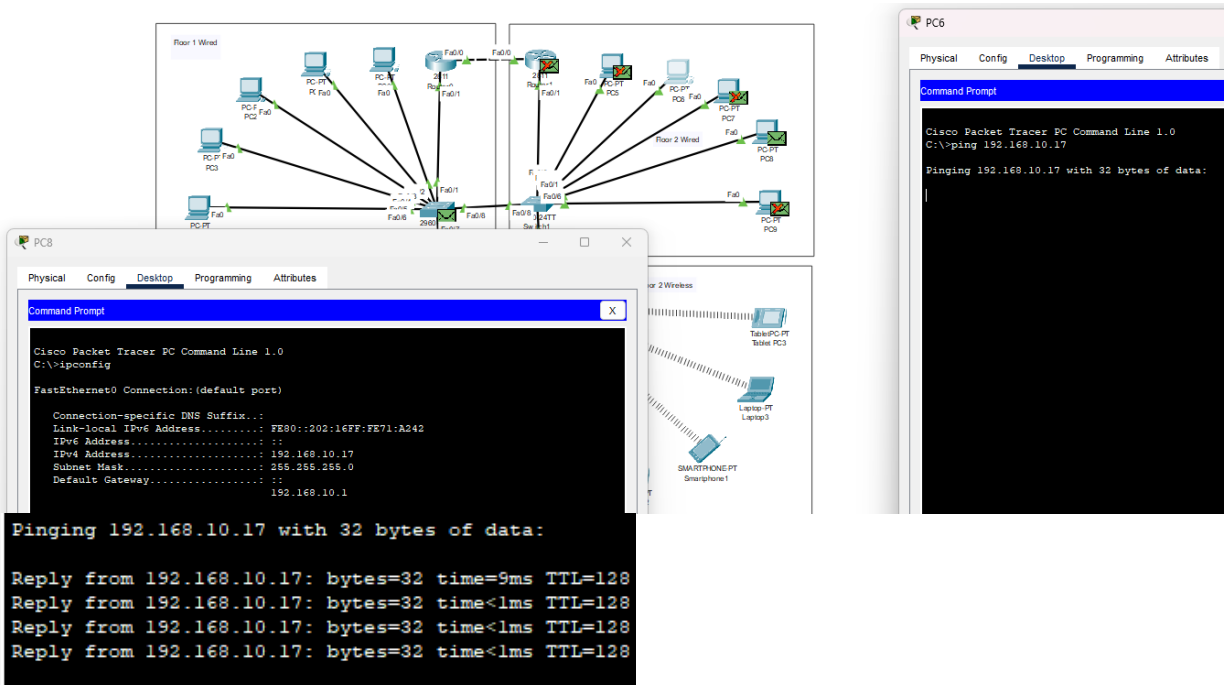
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/8 is directly connected, FastEthernet0/0
L       30.0.0.1/32 is directly connected, FastEthernet0/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, FastEthernet0/1
L       192.168.10.19/32 is directly connected, FastEthernet0/1

R1#
```

Introducing CIDR was very beneficial as this subnet masking allows for the network to divide the IP address into different levels of subnets and with different sizes, this allowed me to create lots of subnetworks under the main domain, without having to waste a great number of IP addresses. The limit of Ips a company can have is no more than 256, this is made for companies who require more, whilst limiting waste

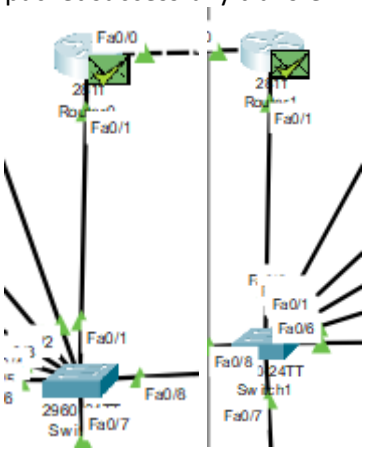
This is beneficial also for scalability as it means if the company was to exceed 256 IP addresses, they would need this Classless Inter-Domain Routing in place which would require a large amount of reconfiguration if we weren't to do this now.

Network Verification



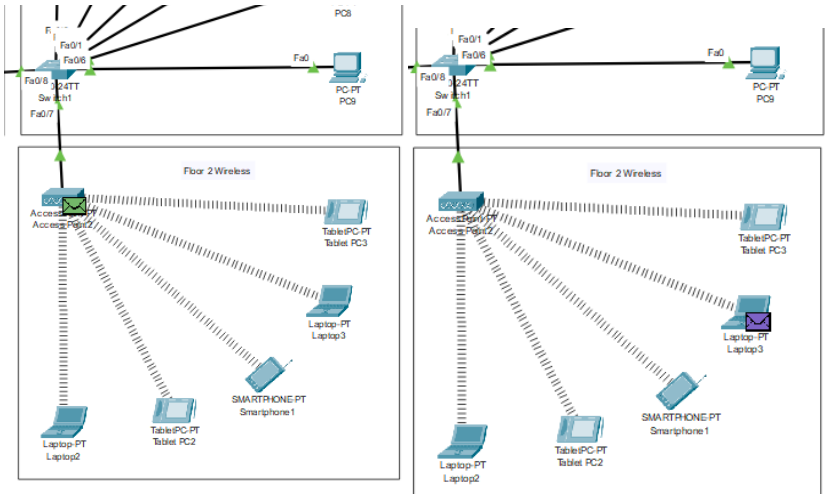
By configurating all my devices properly, I was then yet to test the network, near to the end of the setup. This was done frequently to test individual devices after different changes were made to constantly check for errors before continuing throughout any part of the network design and configuration.

As shown above you can see the ping tests I made from PC6 to PC8, this ping test was done using the simulation tool in cisco packet tracer to identify whether there were any transfer issues throughout each step of communication between the devices. This was necessary to test on the devices to ensure it had no communication errors throughout the development and as you can see this simulation displayed the packet successfully transferring to the switch, and then successfully to device PC8.

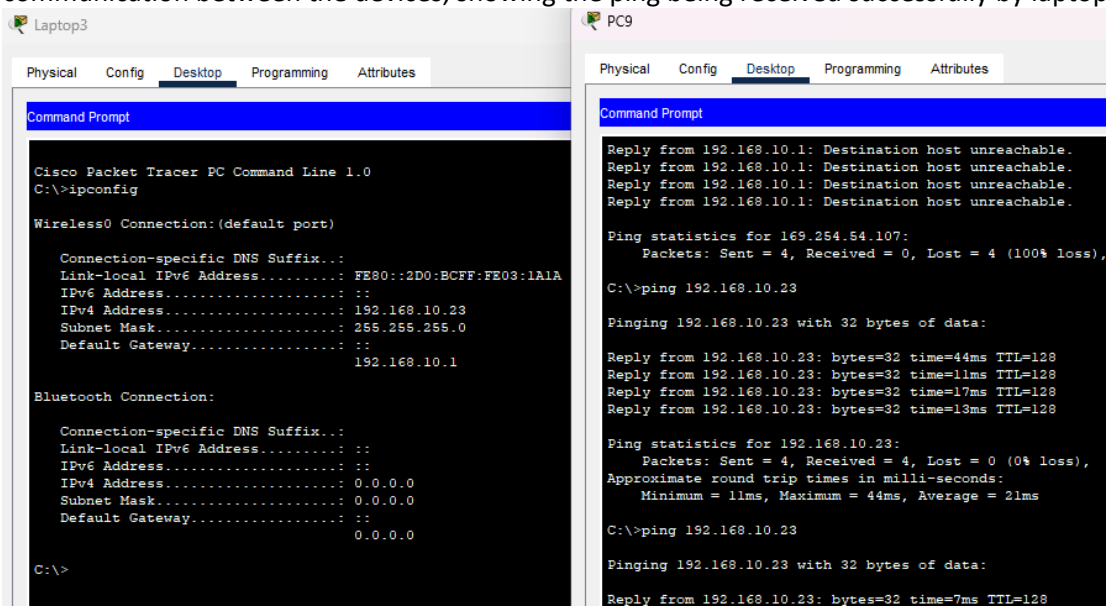


Another verification step I took was to test the communication between the routers and switches to ensure that they can communicate to each other and the other floors. Running the simulation to send packets between the switch and router displayed that the routers and switches were able to send and receive packet transfers successfully in the simulation. Above you can see both routers on the network were able to communicate and receive their transfer messages.

After this, I also tested the ping transfers between the switches and the wireless access points to ensure that the connection is functioning properly and the network can communicate, for example from PC9 to Laptop3



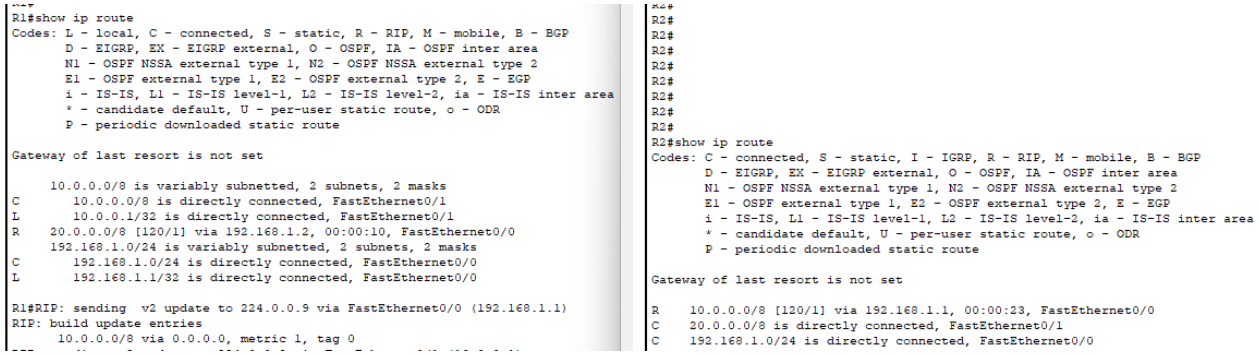
These Images show that the connection between Switch1 and Access Point 2 is working and the communication between the devices, showing the ping being received successfully by laptop3.



After that testing when relayed back to PC9 it shows above the successful reply from the laptop and verifies the network connection and configuration works and all communication between the devices have been verified by performing these tests on all the devices to make sure.

This segment of the report was very important as to meet all the client's requirements I had to ensure that the network was fully functional.

Verifying RIP



```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, FastEthernet0/1
L       10.0.0.1/32 is directly connected, FastEthernet0/1
R    20.0.0.0/8 [120/1] via 192.168.1.2, 00:00:10, FastEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, FastEthernet0/0
L       192.168.1.1/32 is directly connected, FastEthernet0/0

R1#RIP: sending  v2 update to 224.0.0.9 via FastEthernet0/0 (192.168.1.1)
RIP: build update entries
      10.0.0.0/8 via 0.0.0.0, metric 1, tag 0
```

```
R2#
R2#
R2#
R2#
R2#
R2#
R2#
R2#
R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    10.0.0.0/8 [120/1] via 192.168.1.1, 00:00:23, FastEthernet0/0
C    20.0.0.0/8 is directly connected, FastEthernet0/1
C    192.168.1.0/24 is directly connected, FastEthernet0/0
```

The image above shows the verification of the RIP config working after setting up this routing protocol.

After setting up the RIP addresses, we are able to run the command “show ip route” and this will output the IP routes on Router 1 and Router 2 when executed on both. In both command lines it shows the RIP IP used for this protocol as “10.0.0.0”, this verifies the RIP as it shows the code “R” next to it, stating above it means it operating on RIP. The verification above displays the address for the RIP protocol to operate on and tells us that this protocol is running as it should and therefore has the benefits it provides on the network.

Verifying DHCP

```
R1#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.11    0003.E483.7A79          --                      Automatic
192.168.10.16    00E0.F78C.77C1          --                      Automatic
192.168.10.15    0006.2A5B.6846          --                      Automatic
192.168.10.17    000D.BD4E.0039          --                      Automatic
192.168.10.18    0002.16C6.3124          --                      Automatic
192.168.10.19    0002.4A9E.82E6          --                      Automatic
192.168.10.20    0001.C9EE.77DD          --                      Automatic
192.168.10.21    00D0.BAB8.06B0          --                      Automatic
192.168.10.23    00D0.9768.5AD8          --                      Automatic
192.168.10.22    00E0.8F98.AB8B          --                      Automatic
192.168.10.24    0005.5EE4.82C9          --                      Automatic
```

```
R2#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.16    0002.4AE2.AC26          --                      Automatic
192.168.10.11    0001.9713.814C          --                      Automatic
192.168.10.15    0002.1769.D3C6          --                      Automatic
192.168.10.17    0002.1671.A242          --                      Automatic
192.168.10.19    0010.1113.0555          --                      Automatic
192.168.10.20    0030.F2B1.5457          --                      Automatic
192.168.10.21    0005.5E00.A5EC          --                      Automatic
192.168.10.22    0002.4A82.4C36          --                      Automatic
192.168.10.23    00D0.BC03.1A1A          --                      Automatic
192.168.10.24    0005.5EE4.82C9          --                      Automatic
```

To verify the DHCP status we can simply display the IP addresses that are binding to the routers by running “show IP DHCP binding” and this will show all the DHCP IP addresses that are connected to each router when requested. This shows that all the devices across the routers are using the DHCP protocol and working successfully due to successful ping tests showing its connectivity. This verification method is very important as it allows us to make sure that this protocol is running properly and ensures that there are no issues with this protocol and provides better performance for the client.

Evaluation and Discussion

The network design evaluated in this report meets the requirements of the organization by providing a high level of scalability, security, and performance. The chosen hardware and software components have been carefully selected to ensure compatibility and optimal performance, and the configured security measures will protect the network against external and internal threats. In order to carefully select network devices to allow for the client’s needs and potential scalability, I used 2960 switches for each floor to allow for up to 24 wired devices to connect, as well as AP-PT wireless access points to allow for a great number of wireless connections. For security measures, I set up login methods for the network devices to make sure that attackers who may gain access to one of the devices will not be able to log in and run shell commands on the network. This should also be done using Least User Access across the network for minimum access.

Redundancy has been built into the design of the network to ensure that it remains up and running in case of potential planned or unplanned failures/outages. This has been achieved using redundant network devices such as routers and switches, in the network design you can see both switches are connected as a failsafe in case one of the routers encounters issues or downtime. Implementing redundancy protocols such as DHCP, “The major concept is to share the scope of the DHCP server by a

failover system. To load balance and other redundant purposes, the primary DHCP server and the partnering DHCP server can use the same subnetting." (Sanchez & Rahman, no date). With this, it provides failover for both redundancies and balances the load by sharing pools between the DHCP servers.

In addition to the positive aspects of the network design, it's also important I critically evaluated the drawbacks and limitations of the network. One is the cost of the overall network design. Despite the long-term savings and efficiency of the network in terms of scalability and upgrading, it will have a large upfront cost and it is important to consider the organisation's budget. With its complexity, it may also require a higher level of expertise and resources to manage, update and configure when necessary for the future. However, this could be mitigated by installing monitoring and management tools to easily automate, simplify and lower the workload for staff to manage the network, without needing lots of staff to perform this work.

Overall, I believe that this network plan provides a very reliable and efficient solution for them to use and provides great scalability for future purposes. It does, however, create high costs for the company, having to buy these network devices, and set up and have highly trained staff to manage the network due to its complexity. Despite the costs, I believe this will be the best solution for the organisation to operate on and have the options for upgrading and adding devices in the future with its scalability.

Conclusion

In conclusion, the network design and configuration I have presented in this report is a reliable and efficient solution for meeting the organisation's specific needs. During the overall design and setup for this network, I constantly kept in mind the needs of the company and researched and developed the network to be able to create the best and most efficient network design and setup for the company.

In addition, the test and trial of this network and the protocols implemented, outlined in the report, also helped by ensuring that the network is high-performance and fully functional for the needs of the client.

By carefully considering factors like scalability, security and network performance, I have designed a network that will be able to support the company's current and future needs such as upgrading and upscaling. The current security measures and performance configurations will protect the network against threats and improve its performance to be as fast and efficient as possible for a large-scale network with a multitude of devices. Overall, this network design configuration I have made will provide a stable and secure network platform for the company's needs and operation.

References

Ioan Fiţigău, G. T., 2013. *Network performance evaluation for RIP, OSPF and EIGRP routing protocols.* [Online]
Available at: https://ieeexplore.ieee.org/abstract/document/6636217
[Accessed 18 January 2023].
Sanchez, J. G. & Rahman, M. S., no date. *Implementation of DHCP Failover methodology.* [Online]
Available at: https://rahmanmsaidur.com/projects/Fall_16_DHCP_Failover.pdf
[Accessed 23 January 0223].